

Microbiology

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BASICS OF BIOSENSORS AND
MOLECULAR MOTORS

Unit 4

- ▣ Biosensors
- ▣ Types of Biosensors
- ▣ Components of biosensors
- ▣ Medical applications of biosensors
- ▣ Linear motors: actin and myosin
- ▣ Rotatory motors: flagella motor and ATPase

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What is a Biosensor?

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“Biosensor” – Any device that uses specific biochemical reactions to detect chemical compounds in biological samples.
238 aa

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- 1916 First report on immobilization of proteins : adsorption of invertase on activated charcoal
 - 1922 First glass pH electrode
 - 1956 Clark published his definitive paper on the oxygen electrode.
 - 1962 First description of a biosensor: an amperometric enzyme electrode for glucose (Clark)
 - 1969 Guilbault and Montalvo –First potentiometric biosensor:urease immobilized on an ammonia electrode to detect urea
- History of Biosensors

[Page 5]

- 1987 Blood-glucose biosensor launched by MediSense ExacTech
 - 1990 SPR based biosensor by Pharmacia BIACore
 - 1992 Hand held blood biosensor by i-STAT
 - 1996 Launching of Glucocard
 - 1998 Blood glucose biosensor launch by LifeScan FastTake
 - 1998 Roche Diagnostics by Merger of Roche and Boehringer mannheim
 - Current Quantum dots, nanoparticles, nanowire, nanotube, etc
- History of Biosensors

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- 1970 Bergveld -ion selective Field Effect Transistor (ISFET)
 - 1975 Lubbers and Opitz described a fibre-optic sensor with immobilised indicator to measure carbon dioxide or oxygen.
 - 1975 First commercial biosensor (Yellow springs Instruments glucose biosensor)
 - 1975 First microbe based biosensor, First immunosensor
 - 1976 First bedside artificial pancreas (Miles)
 - 1980 First fibre optic pH sensor for in vivo blood gases (Peterson)
 - 1982 First fibre optic-based biosensor for glucose
 - 1983 First surface plasmon resonance (SPR) immunosensor
 - 1984 First mediated amperometric biosensor: ferrocene used with glucose oxidase for glucose detection
- History of Biosensors

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Current Definition

A sensor that integrates a biological element with a physiochemical transducer to produce an electronic signal proportional to a single analyte which is then conveyed to a detector.

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1. LINEARITY Linearity of the sensor should be high for the detection of high substrate concentration.
2. SENSITIVITY Value of the electrode response per

substrate concentration.

3. SELECTIVITY Chemicals Interference must be minimised for obtaining the correct result.

4. RESPONSE TIME Time necessary for having 95% of the response.

Basic Characteristics of a Biosensor

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1. The Analyte A substance of interest that needs detection. For instance, glucose is an 'analyte' in a biosensor designed to detect glucose Protein, toxin, peptide, vitamin, sugar, metal ion.

2. Sample handling Micro fluidics - Concentration increase/decrease, Filtration/selection

Components of Biosensor

3. Bioreceptor A molecule that specifically recognises the analyte is known as a bioreceptor. Enzymes, cells, aptamers, deoxyribonucleic acid (DNA) and antibodies are some examples of bioreceptors. The process of signal generation (in the form of light, heat, pH, charge or mass change, etc.) upon interaction of the bioreceptor with the analyte is termed bio-recognition.

4. Transducer The transducer is an element that converts one form of energy into another. In a biosensor the role of the transducer is to convert the bio-recognition event into a measurable signal. Most transducers produce either optical or electrical signals that are usually proportional to the amount of analyte-bioreceptor interactions.

5. Detection/Recognition How do you specifically recognize the analyte?

6. Signal (How do you know there was a detection)

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Components of a Biosensor
Detector

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Example of biosensors

Pregnancy test

Detects the hCG protein in urine.

Glucose monitoring device (for diabetes patients)

Monitors the glucose level in the blood.

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Example of biosensors

Infectious disease

biosensor from RBS
Old time coal miners'
biosensor

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Research Biosensors
Biacore Biosensor platform

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✓Fluorescence
✓DNA Microarray
✓SPR Surface plasmon resonance
✓Impedance spectroscopy
✓SPM (Scanning probe microscopy, AFM,
STM)
✓QCM (Quartz crystal microbalance)
✓SERS (Surface Enhanced Raman Spectroscopy)
✓Electrochemical
Typical Sensing Techniques
for Biosensors

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Types of Biosensors
1. Calorimetric Biosensor
2. Potentiometric Biosensor
3. Amperometric Biosensor
4. Optical Biosensor
5. Piezo-electric Biosensor

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Calorimetric Biosensors
If the enzyme catalyzed reaction is exothermic,
two thermistors may be used to
measure the difference in resistance
between reactant and product and, hence,
the analyte concentration.

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Potentiometric Biosensor
– For voltage: Change in distribution of charge
is detected using ion-selective electrodes, such
as pH-meters.

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Piezo-Electric Biosensors

The change in frequency is proportional to the mass of absorbed material.

Piezo-electric devices use gold to detect the specific angle at which electron waves are emitted when the substance is exposed to laser light or crystals, such as quartz, which vibrate under the influence of an electric field.

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Electrochemical Biosensors

- For applied current: Movement of e⁻ in redox reactions detected when a potential is applied between two electrodes.

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Optical Biosensors

- Colorimetric for color

Measure change in light adsorption

- Photometric for light intensity

Photon output for a luminescent or fluorescent process can be detected with photomultiplier tubes or photodiode systems.

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Motivated by the application to clinical diagnosis and genome mutation detection

Types DNA Biosensors

- Electrodes
- Chips
- Crystals

DNA biosensor

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Electrochemical DNA Biosensor

- Steps involved in electrochemical

DNA hybridization biosensors:

- Formation of the DNA recognition layer
- Actual hybridization event
- Transformation of the hybridization event into an electrical signal

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Wearable Biosensors

Ring Sensor

Smart Shirt

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Biosensors on the Nanoscale

- ❑ Molecular sheaths around the nanotube are developed that respond to a particular chemical and modulate the nanotube's optical properties.
- ❑ A layer of olfactory proteins on a nanoelectrode react with low-concentration odorants (SPOT-NOSED Project). Doctors can use to diagnose diseases at earlier stages.
- ❑ Nanosphere lithography (NSL) derived triangular Ag nanoparticles are used to detect streptavidin down to one picomolar concentrations.
- ❑ The School of Biomedical Engineering has developed an anti-body based piezoelectric nanobiosensor to be used for anthrax, HIV hepatitis detection.

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- ❖ Food Analysis
 - ❖ Study of biomolecules and their interaction
 - ❖ Drug Development
 - ❖ Crime detection
 - ❖ Medical diagnosis (both clinical and laboratory use)
 - ❖ Environmental field monitoring
 - ❖ Quality control
 - ❖ Industrial Process Control
 - ❖ Detection systems for biological warfare agents
 - ❖ Manufacturing of pharmaceuticals and replacement organs
- Application of Biosensor

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Potential Applications

- Clinical diagnostics
- Food and agricultural processes
- Environmental (air, soil, and water) monitoring
- Detection of warfare agents.

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- Biosensors play a part in the field of environmental quality, medicine and industry mainly by identifying material and the degree of concentration present

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- Molecular Machines/Motors
- Properties of ATP-based Protein Molecular Machines
- The F₀F₁-ATP Synthase Motors
- Coupling and Coordination of Motors
- The Bacterial Flagellar Motor
- Flagellar motor structure

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Molecular Machines/Motors

- Molecular machines can be defined as devices that can produce useful work through the interaction of individual molecules at the molecular scale of length.
- A convenient unit of measurement at the molecular scale would be a nanometer. Hence, molecular machines also fall into the category of nanomachines.
- As our knowledge and understanding of these numerous machines continues to increase, we now see a possibility of using the natural machines, or creating synthetic ones from scratch, by mimicking nature.

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Different types of molecular machines. These machines used by nature for force generation and motion. They convert chemical energy into mechanical force via conformational changes.

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- Most of the natural machinery is built from proteins.
- These findings help unravel the mysteries associated with the molecular machinery and pave the way for the production and application of these miniature machines in various fields, including medicine, space exploration, electronics and military.
- We divide the molecular machines into three broad categories—protein based, DNA-based, and chemical molecular motors.

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Properties of ATP-based Protein Molecular Machines

- Two form the F₀F₁-ATP synthase, and the third one is the bacterial flagellar motor.
- The protein-based molecular motors rely on an energy-rich molecule known as adenosine triphosphate (ATP),
- The machines described in this section, the F₀F₁-ATPase, the kinesin, myosin, and dynein superfamily of protein molecular machines, and bacteria flagellar motors all depend, directly or indirectly, on ATP for their input energy.
- These machines have now been segregated out of their natural environment and are seen as energy conversion devices to obtain forces, torques, and motion.
- One disadvantage associated with ATP dependence is that the ATP creation machinery itself could be many times heavier and bulkier than the motors, thereby making it more complex.
- These machines perform best in their natural environment, and in the near future it may be possible to have them as a part of biomimetic molecular machinery.

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The F₀F₁-ATP Synthase Motors

- ATP is regarded as the energy currency of biological systems.
- When this currency is utilized (i.e., the energy of the molecule that is used to drive a biological process), the terminal anhydride bond in the ATP molecule has to be split. This leaves adenosine diphosphate (ADP) and a phosphate ion (Pi) as the products, which are recombined to form ATP by a super efficient enzyme motor assembly called the F₀F₁-ATP synthase (F₀F₁-ATPase).
- ATP synthase is present inside the mitochondria of animal cells, in plant chloroplasts, in bacteria, and some other organisms.

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- ATP synthase consists of two portions: a membrane-spanning portion, F_o, comprising the ion channel, and a soluble portion, F₁, containing three catalytic sites.
- Both F_o and F₁ are reversible rotary motors --- perhaps the smallest motors known to science. F_o uses the transmembrane electrochemical gradient to generate a rotary torque to drive ATP synthesis in F₁ or, when driven backwards by the torque generated in F₁, to pump ions uphill against their transmembrane electrochemical gradient.

- F1 generates a rotary torque by hydrolyzing ATP at its three catalytic sites or, when turned backwards by the torque generated in Fo, as a synthesizer of ATP.

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The F₀F₁-ATPase motors. The F₀ motor is embedded in the inner mitochondrial membrane of the mitochondria. F₀ is typically composed of a, b, and c subunits as shown. The F₁ motor is the soluble region composed of three α-, three β-, one each of γ -, δ- and ε-subunits.

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Coupling and Coordination of Motors

- The ATP synthase is actually a combination of two motors functioning together, the hydrophobic transmembrane F₀-ATPase motor and the globular F₁-ATPase motor

The F₀F₁-ATPase contains two rotary motors: the membrane -bound F₀, driven by the flux of ions across a membrane, and the soluble F₁, driven by ATP hydrolysis. These two motors are coupled by sharing a common motor, drawn in white, and a common stator, drawn dark. The figure shows the F₀F₁-ATPase operating as an ATP-synthase. Rotation of F₀ driven by ion flux drives F₁ in reverse, causing it to synthesize ATP from ADP and inorganic phosphate.

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The Bacterial Flagellar Motor

- A flagellum (plural: flagella) is a long, slender projection from the cell body, composed of microtubules and surrounded by the plasma membrane.
- In small, single-cell organisms they may function to propel the cell by beating in a whip-like motion; in larger animals, they often serve to move fluids along mucous membranes such as the lining of the trachea.
- The bacterial flagellar motor is a nanotechnological marvel, no more than 50 nm in diameter, built from about 20 different kinds of parts.

Examples of bacterial flagella arrangement schemes. A-

Monotrichous; B-Lophotrichous;
C-Amphitrichous; D-Peritrichous

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- It spins clockwise (CW) or counterclockwise (CCW) at speeds on the order of 100 Hz, driving long thin helical filaments that enable cells to swim.
- Peritrichously flagellated cells (peri, around; trichos, hair), such as *Escherichia coli*, execute a random search, moving steadily at about 30 diameters per second, now in one direction, now in another.
- Steady motion requires CCW rotation. Receptors near the surface of the cell count molecules of interest (sugars, amino acids, dipeptides) and control the direction of flagellar rotation.
- If a leg of the search is deemed favorable, it is extended, i.e., the motors spin CCW longer than they otherwise would. This bias enables cells to actively find regions in their environment where life is better.

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- Thus, the flagellar motor is the output organelle of a remarkable sensory system, the components of which have been honed to perfection by billions of years of evolution.
- A number of bacterial species in addition to *E. coli* depend on flagella motors for motility: e.g., *Salmonella enterica* serovar, Typhimurium (*Salmonella*), *Streptococcus*, *Vibrio* spp., *Caulobacter*, *Leptospira*, *Aquaspirillum* *serpens*, and *Bacillus*.
- The rotation of flagella motors is stimulated by a flow of ions through them, which is a result of a build-up of a transmembrane ion gradient. There is no direct ATP-involvement; however, the proton gradient needed for the functioning of flagella motors can be produced by ATPase.

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Flagellar motor structure

- Bacterial flagella are the only biological structures known that use rotation for the purpose of locomotion.
- Flagella consist of a rotary motor embedded in the cell envelope connected to an extracellular helical propeller. The motor is powered by the flow of ions down an electrochemical gradient across the cytoplasmic membrane into the cell.
- The ions are typically H^+ (protons), although certain marine and alkalophilic species have motors driven by Na^+ .
- The electrochemical gradient (protonmotive force or sodium motive force) consists of a transmembrane voltage and a concentration difference across the membrane, both of which are maintained by various metabolic processes.
- The rotor shown in white (Figure 5), consists of a series of rings spanning the cell envelope and is attached via the flexible hook to the helical propeller or filament. The stator is a ring of particles in the cytoplasmic membrane, containing the proteins MotA and MotB, and anchored to the peptidoglycan cell

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The bacterial flagellar motor. The rotor, drawn in white, consists of a series of rings that span the cell envelope and are attached to the extracellular hook and filament. The stator consists of a ring of torque-generating units containing the proteins MotA and MotB and anchored to the cell wall. Ions flowing through the motor generate torque by means of unknown interactions between the rotor and stator in the vicinity of the C-ring. The F₀F₁-ATPase is also shown, to the same scale.

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The fuel/air intake system red, the electrical system green, and the exhaust system blue

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Cytoskeleton: Microtubules, Microfilaments, and Intermediate filaments

- Cytoskeleton gives the cell shape, structure, and physical organization
- Motor proteins can rearrange the structural elements, or move cell components around the cytoskeleton.
- The cytoskeleton is in constant state of change depending on the

requirements of the cell.

- 3 major structural elements of the cytoskeleton

- Microtubules - hollow, rigid cylindrical tubes made from tubulin subunits

- Microfilaments - solid, thinner structures made of actin

- Intermediate filaments - tough, ropelike fibers made of a variety of related proteins

Mitochondrion

Plasma membrane

Ribosomes

Rough

Endoplasmic

reticulum

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The Myosin Linear Motor

- Myosin is a diverse superfamily of motor proteins.
- Myosin-based molecular machines transport cargoes along actin filaments, the two-stranded helical polymers of the protein actin that are about 5-9 nm in diameter. They do this by hydrolyzing ATP and utilizing the released energy.
- In addition to transport, they are also involved in the process of force generation during muscle contraction, wherein thin actin filaments and thick myosin filaments slide past each other.
- Myosin molecules were first seen (in the late 1950s) through electron microscope protruding out from thick filaments and interacting with the thin actin filaments. Since that time, ATP has been known to play a role in myosin-related muscle movement along actin; however, the exact mechanism was unknown, until it was explained in 1971.

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Myosin cross-bridge model of contraction. Myosin uses the energy released from ATP hydrolysis to move along actin.

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The Kinesin Linear Motor

- The kinesin and dynein families of proteins are involved in cellular cargo transport along microtubules, in contrast to myosin, which transports along actin.

- Microtubules are 25-nm diameter tubes made of protein tubulin and are present in the cells in an organized manner.
- Microtubules have polarity; one end being the plus (fast-growing) end while the other end is the minus (slow-growing) end.
- Kinesins move from the minus end to the plus end of the microtubule, whereas dyneins move from the plus end to the minus end.
- Microtubule arrangement varies in different cell systems. In nerve axons, they are arranged longitudinally such that their plus ends point away from the cell body and into the axon.
- In epithelial cells, their plus ends point toward the basement membrane.
- They extend radially out of the cell center in fibroblasts and macrophages with the plus end protruding outward.
- Similar to myosin, kinesin is also an ATP-driven motor.
- One unique characteristic of the kinesin family proteins is their processivity; they bind to microtubules and literally walk on it for many enzymatic cycles before detaching.

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The Dynein Motor

- The dynein superfamily of proteins was discovered in 1965.
- Dyneins exist in two isoforms: cytoplasmic and axonemal.
- Cytoplasmic dyneins are involved in cargo movement, whereas axonemal dyneins are involved in producing bending motions of cilia and flagella.
- Because dynein is a larger and more complex structure than other motor proteins, its mode of operation is not as well known. However, electron microscopy and image processing was used to show the structure of a flagellar dynein at the start and end of its power stroke, which gives some insight into its possible mode of force generation.

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Cilia and flagella have a core axoneme, a complex of microtubules and associated proteins.

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Metals

Semiconductor

Materials

Ceramics

Polymers

Synthetic

BIOMATERIALS

Orthopedic

screws/fixation

Dental Implants Dental Implants

Heart

valves

Bone

replacements

Biosensors

Implantable

Microelectrodes

Skin/cartilageDrug Delivery

Devices Ocular

implants

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Biomaterial Science

[Page 3]

Bileaflet Heart Valves

[Page 4]

Dental Implants

[Page 5]

Intraocular Lenses

□

Implantation often performed on
outpatient basis.

[Page 6]

Vascular Grafts

□

Must Be Flexible.

□

Designed With Open
Porous Structure.

□

Often Recognized By

Body As Foreign.

[Page 7]

Hip

-

Replacements

[Page 8]

Normal versus Arthritic Hip

Normal Hip: note the space
between the femur and
acetabulum, due to cartilage
Arthritic Hip: No space
visible in joint, as
cartilage is missing

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using a Polymethylmethacrylate
(PMMA) cement to adhere the
metal to the bone
using a porous metal
surface to create a bone
ingrowth interface
Two attachment methods

[Page 10]

the acetabulum and the proximal
femur have been replaced. The
femoral side is completely metal.
The acetabular side is composed
of the polyethylene bearing
surface
Overview of femoral
replacement

[Page 11]

The two materials are
bonded and equal force is
applied to both
Load transfer in Composite materials

[Page 12]

Comparison: Moduli of Elasticity

Modulus of elasticity of different implant materials and bone (in GPa)

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Implant bonding

A bonded interface is
characteristic of a
cemented prosthesis
(left)

non-bonded interface is
characteristic of a non-
cemented press fit
prosthesis (right)

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Degradation Problems

Example of fractured
artificial cartilage from
a failed hip
replacement

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Chapter 4

Biomaterials: Basic principles

4.1 Introduction

Human tissues and organs sometimes fail to perform their regular actions due to genetic defects, age, illness, trauma, degeneration, or injuries. Some of these conditions are handled with the use of day-to-day medication, i.e., drugs. However, some cannot be repaired/rectified by providing medicines and require the use of unique materials and devices. These then bring about the inevitability of surgical repair, which encompasses anatomical sections such as knee joints, elbow joints, vertebrae, teeth, and other crucial organs such as heart, skin, kidney, etc. In the broadest sense, the unique materials (other than drugs) or combinations of materials that are primarily expected to be used inside a mammal or human to treat, repair, augment or replace any tissue is referred to as biomaterials. These biomaterials can either be formed from nature or physically manufactured by utilizing a wide range of physical and chemical methods. The field of biomaterials is multidisciplinary (Figure 4.1). The design of a simple biomaterial (for example, a bone screw or bone plate) neces -

sitates knowledge and ideas from multiple disciplines. It needs the synergistic integration of materials, biology, medicine, mechanical sciences, and chemistry. The number of medical devices used annually by humans is significant, estimated at 1.5 million by the World Health Organization, with around 10,000 forms of standardized device classes available worldwide.

The term “biomaterials” is often described as any material that comes into contact with humans or animals to fulfil their intended function without causing any toxic reaction. This is the single most crucial aspect that differentiates a biomaterial from any other material, i.e., its capacity to be in

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contact with human body tissues without instigating an undesirable degree of response. For thousands of years, humans have inevitably used or at least attempted to use materials to make devices and instruments of practical value by converting basic available substances into materials. Biomaterials have a long history of medical use, and at various times they were seen in a different way. Based on the development and its uses, the broad definition has evolved over the years and could be further expanded as new applications in medicine emerge. Nevertheless, the meaning of biomaterials attained a harmony of opinion by scientists worldwide and is now defined now as “a material designed to take a form that can direct, through interactions with living systems, the course of any therapeutic or diagnostic procedure.” In most cases, a biomaterial is any natural or synthetic biocompatible material that is used to replace or assist part of an organ or a tissue while maintaining close contact with living tissue. They may be produced from

Figure 4.1: Interdisciplinary system of biomaterials (Cooke et al., 1996; Marin et al., 2020; Ramakrishna et al., 2010; Ratner, 1996).

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materials including solid, liquid, and gel substances (metallic components, polymers, ceramics or composite materials). It should be noted that the bio prefix of biomaterials applies to biocompatible, rather than biological or biomedical, as many would intuitively assume.

Biomaterials, a fascinating and highly interdisciplinary area, have grown to become an integral component in the modern-day improvement of human condition and quality of life. It is accomplished by addressing numerous health-related issues that come from several sources. Over the past few decades, biomaterials have broadened their applications from diagnostics (gene arrays and biosensors) and medical equipment (blood bags, surgical tools) to therapeutic medications (medical implants and devices) and emerging regenerative drugs (tissue-engineered skin and cartilage), and more. Many uses of synthetic and naturally occurring medicinal materials can be classified by their place and role in the human body: skeletal system (joint replacements, bone fractures and defects, artificial tendons and ligaments, dental implants), cardiovascular system (blood vessel prosthesis, heart valves), organs (artificial heart, skin repair), etc. The vital applications are presented in Figure 4.2 and tabulated in Table 4.1. Figure 4.2: Biomaterials have made an enormous impact on the treatment of injury and disease and are used throughout the body (Cooke et al., 1996; Kuhn, 2005; Ratner et al., 2020).

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Table 4.1: Applications of biomaterials.

Organ/Tissue Examples

Heart Biventricular pacemakers, artificial valves, artificial heart

Eye Contact lens, artificial intraocular lens

Ear Hair-clip artificial stapes, cochlear implants

Muscle Sutures

Kidney Dialysis machines

Skin Skin wound repair, burn dressings, skin substitutes, artificial skin

Circulation Synthetic blood vessels

Bone Bone screws, plates, pins, intramedullary rods, bone cement to repair defects

Teeth Fillings and replacements

4.2 Desired properties of biomaterials

As discussed earlier, the ultimate goal of biomaterials in medicine is to treat, improve, or substitute tissue organs (e.g., bone, muscle, skin) or body function. Such goals can be accomplished by integrating the properties of materials, nature of the system, device architecture, and physiological specifications. The biomaterial application method must combine the chemical and mechanical features of the biological system in order to achieve the required functional results (Figure 4.3). For the processing of biomaterials and related medical items, different elements from science and social parameters are also considered.

4.2.1 Biocompatibility

From a biological point of view, biocompatibility is nothing but the acceptability of non-living materials (synthetic or natural) in a living body (mammal/human). This acceptance is broadly defined as the “ability of a material to perform its desired functions with respect to a medical therapy, to induce an appropriate host response in a specific application and to interact with living systems without having any risk of injury, toxicity, or rejection by the immune system and undesirable or inappropriate local or systemic effects”. Since a biomaterial is intended to be used in intimate contact with living tissues, it is important that the implanted material does

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not cause any undesired reactions to the surrounding tissues and host organs. Apart from that, the implanted material is expected not to subdue the activity of normal cells or provoke any unwanted local or systemic reactions in the recipient, but is encouraged to generate positive cellular or tissue response.

Usually, the criteria for this biocompatibility are dynamic and detailed and differ with individual specific medical applications. For example, a specially designed material (screw or plate) may possibly be biocompatible in bone implant surgery, but the same components may not be biocompatible in skin applications.

4.2.2 Host response

Host response is defined as the “response of the host organism (local and systemic) to the implanted material or device”. Most of the materials are never inert, and a biomaterial’s clinical success depends significantly on the host tissue’s reaction with the foreign material. For material establishment-host tissue interaction, these reactions depend heavily on the time-

length, purpose, and site of implantation.

4.2.3 Non-toxicity

Toxicity of biomaterials deals with elements that migrate out of the bio - materials. A carefully designed biomaterial should serve its purpose in the Figure 4.3: Primary requirements of new material design.

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living body's environment without negatively influencing other cells, organs, or the whole organism. It is logical to assume that, unless specifically designed to do so, a biomaterial should not release or generate anything from its bulk.

4.2.4 Mechanical properties

In addition to biocompatibility, mechanical properties are embedded in biomaterial design prior to implementation and will majorly contribute to the outcome. Materials undergo several forces which are primarily stress, strain, shear, and a mixture of these. The tensile strength, yield strength, elastic modulus, corrosion, creep, and hardness are some of the most important properties of biomaterials that should be carefully studied and evaluated before implantation. For hard tissue applications, the mechanical properties are of top priority.

4.2.5 Corrosion, wear, and fatigue properties

Wear is often considered as one of the leading causes of implant failure. In some cases, wear has also been shown to accelerate the corrosion of biomedical devices and implants. The fatigue resistance is related to the material's reaction to repeated cyclic loads.

4.2.6 Design and manufacturability

Appropriate material design is also one of the critical factors to consider for biomaterials. For example, in the early 1900s, bone plates were utilized to assist in fixing long bone fractures. Many of these experimental plates broke because of their primitive mechanical design where they were too narrow and had a stress-focusing corner. Manufacturability is the ability to manufacture the item with relative ease that is ideal for its intended use at minimal cost and high reliability. Concerning biomaterials, the manufacturability, in a broader sense, incorporates the potential of the material to be sterilized by a validated sterilization technique which is deemed appropriate for biomedical applications. The material is

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expected to not get impaired by standard sterilizing techniques such as autoclaving, dry heat, radiation, ethylene oxide, etc. (see Chapter 8 for more details).

In summary, for any material to qualify as a biomaterial, it should

- Be biocompatible, i.e., non-toxic, non-carcinogenic, non-allergenic, etc.
- Have required/suitable physical and chemical properties
- Have suitable mechanical properties
- Have stable durability for the period it is intended for, i.e., hours to years
- Be easy to process with the available techniques
- Be sterilizable with current facilities without any difficulty
- Be cost-effective and accessible

4.3 Types of biomaterials

Depending on the chemical bonding, materials can be categorized into

three broad groups. These are (i) polymers, (ii) ceramics, and (iii) metals.

Because these material structures vary by their nature of bonding (covalent, ionic or metallic), they have different properties, and hence different uses in the body (Table 4.2). Different kinds of biomaterials and chemical bonds associated with it are shown in Figure 4.4. Metals and ceramics are differentiated in the field of materials engineering, although they are both Figure 4.4: Different kinds of biomaterials and chemical bonds associated with them (Gul et al., 2020; He et al., 2017; Zindani et al., 2019).

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Table 4.2: Classification of biomaterials and typical properties associated with them (Wagner, 2020).

Attributes Polymers Metals & Alloys Ceramics

Type of bonds

present

Covalent & van der

Waals forces

Metallic Ionic/Covalent

Melting point Low Intermediate High

Chemical stability Poor Good Very high

Electrical

conductivity

Very low High Very low, but varies

Thermal

conductivity

Very low to

intermediate

High Low

Properties and

advantages

Degradable, inert,

similar density to

soft tissues and ease

of processing

High strength and

hardness

Non-conductive and

inert; closely

mimic biological

properties of bone

Mechanical

deformation

Very high, plastic (can

be easily shaped and

processed)

High (ductile) Low (brittle)

Major issues Thermally unstable;

low strength

Wear and

corrosion

High density and

brittle

Biomedical

applications

Soft tissue implants;

drug delivery

systems; tissue

engineering

Hard tissue

applications

(Orthopedic and
dental implants)

Tissue engineering

inorganic compounds. Each material has its own benefits and drawbacks, and its biomedical applications are decided according to its individual properties and intended place substitution.

4.3.1 Metals

Metals are the most widely known biomedical materials and are indispensable in the medical field. Nearly all of the metal biomaterials are crystalline in nature, i.e., with regular atomic arrangements. Metals have great strength, resistance to fracture toughness, better elasticity, and rigidity compared with ceramics and polymers. Their outstanding

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mechanical reliability properties are controlled by dislocation and crystallization. For this reason, metals are extensively employed for load-bearing implant applications such as orthopaedic, dental, and maxillofacial surgery. Apart from these, metals are also used in making stents and stent-grafts for cardiovascular surgeries. The most common metals and alloys used for biomedical applications are stainless steel, titanium, titanium-based alloys, cobalt-based alloys, magnesium-based alloys, and tantalum-based alloys.

4.3.2 Ceramics

Ceramics are inorganic solid materials consisting of metallic and non-metallic elements that are predominantly bound together by ionic bonds. They exist as both crystalline and non-crystalline (amorphous) compounds. Ceramics are typically characterized by excellent biocompatibility, high corrosion resistance, high wear, high strength, extremely high stiffness, and hardness. The advancement of ceramic material applications in the biomedical industry has focused mostly on orthopaedics and dentistry. Bioinert ceramics such as alumina (Al_2O_3), zirconia (ZrO_2) and pyrolytic carbon, and bioresorbable ceramics such as calcium phosphates are some of the widely employed bioceramics.

4.3.3 Polymers

Polymers are macromolecules and they represent a major and versatile

class of biomaterials being widely applied in biomedical applications due to their low toxicity in biological fluids, easy pre/post-processing, sterilization, better shelf life, lightweight nature, and remarkable physical and chemical properties.

Biomaterial use is primarily dependent on the need, function, and environment of the intended application. In a variety of implants, metal, polymers, ceramics, and composites are being used. The specific material or material combinations for use in a device will significantly affect both its patient performance and marketing potential. The next three chapters will discuss in detail each of the material types.

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4.4 Multiple choice questions

1. Biomaterials

- (a) are always synthetic materials; natural materials are not employed
- (b) are always natural materials; synthetic materials are not employed
- (c) can be natural or synthetic materials
- (d) are always polymeric materials

2. Select the statement which correctly relates to biocompatibility.

- (a) a biocompatible material should provide healing characters
- (b) a biocompatible material should have therapeutic characteristics
- (c) a material is considered as a biocompatible material as long as it causes no harm to the host body
- (d) a biocompatible material should have the exact dimensions as the damaged tissue or part

3. Select the option(s) which do(es) NOT come under the class of bio-compatible materials.

- (a) eyeglasses; wheelchair
- (b) contact lenses; dental implants
- (c) orthopedic implant; stents
- (d) external hip prosthesis; massage footwear

4. Which of the following has the best osteointegration properties?

- (a) SS316
- (b) porous titanium
- (c) Co-Cr alloys
- (d) all of the above

5. Which of the following is/are biomaterial(s)?

- I. materials used for tooth filling
- II. materials used for cardiovascular repairs
- III. glucose meters and stethoscopes
- IV. materials used for hip implants

- (a) only I
- (b) only I, II and IV
- (c) only I, III and IV
- (d) only III

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6. Select the option(s) which is/are TRUE about biodegradation.

- (a) it depends on the molecular architecture
- (b) it is a precise breakdown of the material over time
- (c) metals biodegrade faster than polymeric materials
- (d) all of the above

7. Which class of biomaterials has chemical structures similar to bone?

- (a) polymeric biomaterials
- (b) ceramic biomaterials
- (c) metallic biomaterials
- (d) all of the above

8. Which class of biomaterials encourages bonding with surrounding
tissues and stimulates new bone growth?

- (a) bioinert ceramics
- (b) bioactive ceramics
- (c) Co-Cr alloys
- (d) all of the above

9. Which of the following is TRUE?

- (a) ceramics possess excellent wear and friction properties
- (b) SS316, Co-Cr and Ti alloys form a protective oxide layer on their
surfaces
- (c) bioceramics are more reactive than certain metallic implants
- (d) all of the above

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